03_dirty_read_demo

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

Demonstrates the risk of reading uncommitted data.

- Requires two sessions running simultaneously.
- SQL Server 2019+ | Database: STN_Lab

3.1 SETUP Create an isolated test record to track clearly

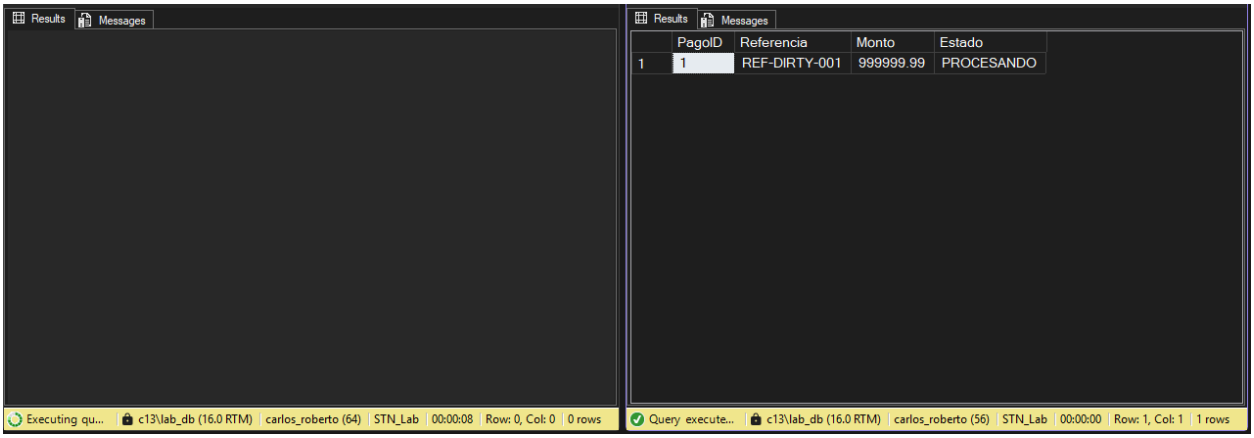
```
USE STN_Lab;
GO

-- ============================================================
==
-- SETUP — Create an isolated test record to track clearly
-- ============================================================
==
IF OBJECT_ID('dbo.PagoAuditTest', 'U') IS NOT NULL DROP TABLE
dbo.PagoAuditTest;
GO

CREATE TABLE dbo.PagoAuditTest (
    PagoID       INT           NOT NULL IDENTITY(1,1) PRIMARY
KEY,
    Referencia   VARCHAR(50)   NOT NULL,
    Monto        DECIMAL(18,2) NOT NULL,
    Estado       VARCHAR(20)   NOT NULL DEFAULT 'PENDIENTE'
```

```sql
);
GO

INSERT INTO dbo.PagoAuditTest (Referencia, Monto, Estado)
VALUES ('REF-DIRTY-001', 50000.00, 'PENDIENTE');
GO

-- Confirm initial state
SELECT * FROM dbo.PagoAuditTest;
-- Expected: Monto = 50000.00, Estado = PENDIENTE
```

3.2 Simulates a payment processing transaction

```sql
-- SESSION A — Run this first and LEAVE IT OPEN

-- ====================================================

BEGIN TRANSACTION;

    UPDATE dbo.PagoAuditTest
    SET Monto  = 999999.99,
        Estado = 'PROCESANDO'
    WHERE Referencia = 'REF-DIRTY-001';

    -- DO NOT COMMIT — the transaction is "in flight"
    -- Wait while Session B reads the dirty data
    WAITFOR DELAY '00:00:30';

-- *** Session A will ROLLBACK after 30 seconds ***
ROLLBACK;
-- The update never happened. Monto returns to 50000.00.

-- ====================================================
-- SESSION B — Run WHILE Session A is open
-- ====================================================
```

```sql
-- BAD PRACTICE: Read Uncommitted reads dirty (uncommitted) d
ata
SET TRANSACTION ISOLATION LEVEL READ UNCOMMITTED;

SELECT
    PagoID,
    Referencia,
    Monto,
    Estado
FROM dbo.PagoAuditTest
WHERE Referencia = 'REF-DIRTY-001';
```

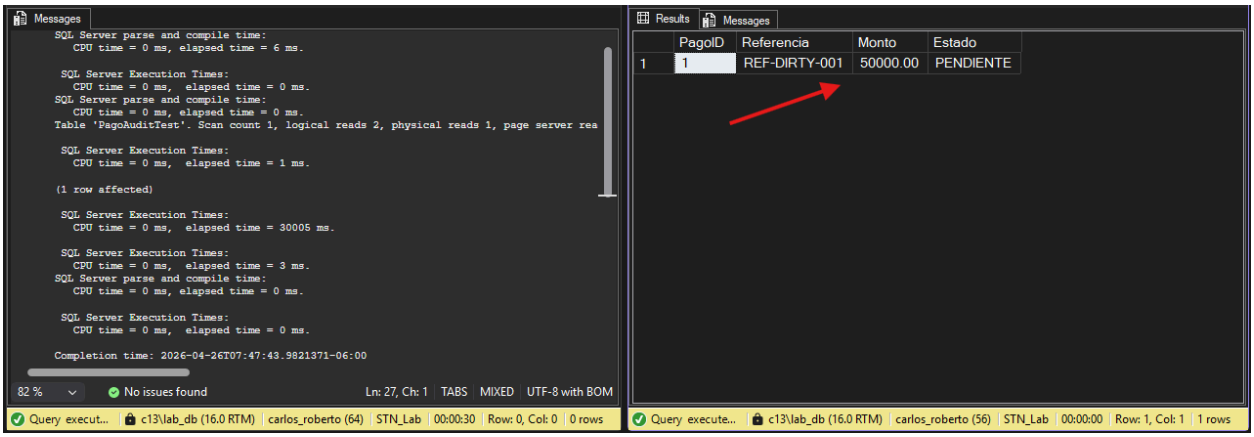


OBSERVED:

- Monto = 999999.99 is DIRTY DATA, uncommitted
- Estado = 'PROCESANDO' also dirty

This when transaction is reverted:

- Then re-run the same SELECT: Monto returns to 50000.00 — the 999999.99 never existed



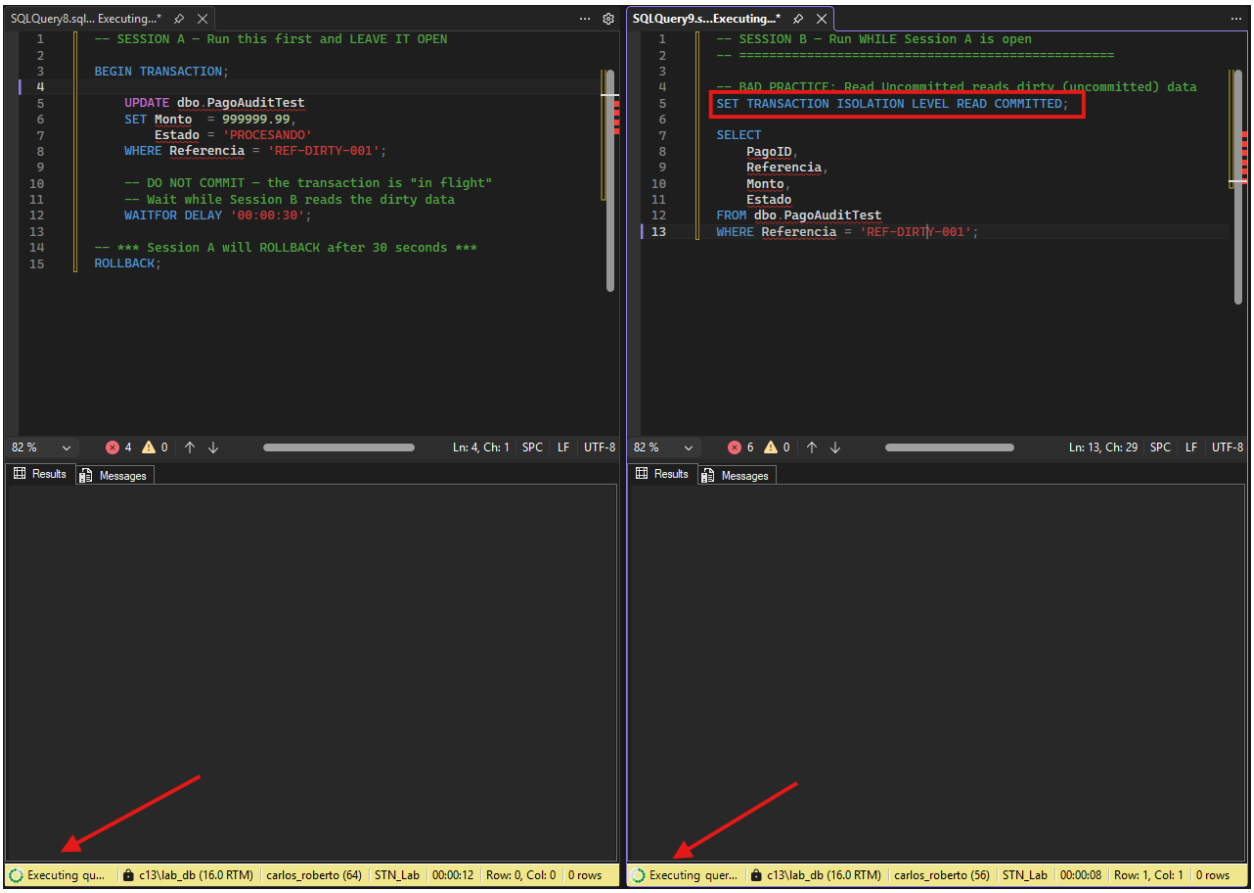
3.3 ILLUSTRATION OF THE REAL RISK IN STN_Lab

Scenario:

A fiscal auditor runs a report using NOLOCK (= READ UNCOMMITTED) to avoid blocking. The batch job is recalculating 200,000 declarations.

- The auditor's report captures mid-flight values.
- The batch rolls back due to a constraint violation.
- The auditor's report shows fiscal totals that NEVER EXISTED in the database.
- Result: regulatory report based on ghost data.

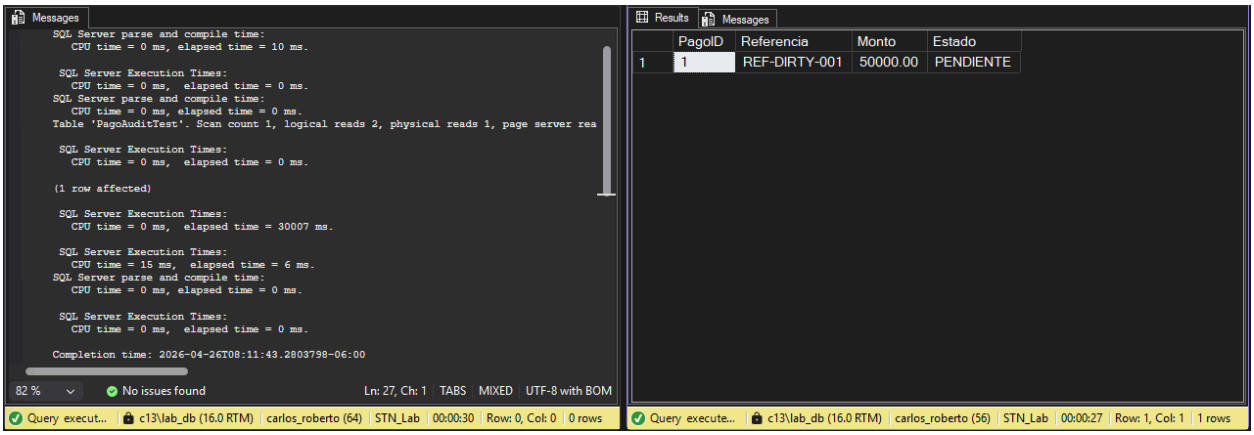
SAFER ALTERNATIVE: Use default Read Committed



OBSERVED:

Query waits (blocks) until Session A commits or rolls back.

Once Session A rolls back: Monto = 50000.00 correct committed value.



3.4 CLEANUP

```sql
DROP TABLE IF EXISTS dbo.PagoAuditTest;
GO
```